

SAFETY DATA SHEET

Sodium Hydroxide Solution, 50%

According to OSHA Hazard Communication Standard, 29 CFR 1910.1200

Revision date: March 18, 2024 · Version 4.1 · Supersedes version 4.0 (August 2, 2021)

SECTION 1. IDENTIFICATION

Product identifier: Sodium Hydroxide Solution, 50% (w/w)

Synonyms: Caustic soda solution, lye solution, NaOH 50%

Recommended use: Industrial pH adjustment, cleaning, chemical manufacturing

Supplier: Northwest Alkali Products, Inc., 7200 N Lombard Street, Portland, Oregon 97203 · Telephone (503) 555-0140

Emergency telephone: CHEMTREC 1-800-424-9300 (24 hours)

SECTION 2. HAZARD IDENTIFICATION

Classification: Skin corrosion, Category 1A (H314); Serious eye damage, Category 1 (H318); Corrosive to metals, Category 1 (H290)

Signal word: DANGER

Hazard statements: H290 May be corrosive to metals. H314 Causes severe skin burns and eye damage.

Precautionary statements: P260 Do not breathe mist or vapours. P280 Wear protective gloves, protective clothing, eye protection and face protection. P301+P330+P331 If swallowed: rinse mouth, do NOT induce vomiting. P303+P361+P353 If on skin (or hair): take off immediately all contaminated clothing, rinse skin with water. P305+P351+P338 If in eyes: rinse cautiously with water for several minutes; remove contact lenses if present and easy to do; continue rinsing. P310 Immediately call a poison center or doctor. P390 Absorb spillage to prevent material damage. P406 Store in a corrosion-resistant container with a resistant inner liner.

Pictograms: Corrosion

SECTION 3. COMPOSITION / INFORMATION ON INGREDIENTS

Sodium hydroxide, CAS 1310-73-2, 48–52%. Water, CAS 7732-18-5, 48–52%.

SECTION 4. FIRST-AID MEASURES

Eyes: Immediately flush with plenty of water for at least 30 minutes, lifting the eyelids. Get medical attention immediately.
Skin: Immediately flush with plenty of water for at least 30 minutes while removing contaminated clothing and shoes. Get medical attention immediately.
Inhalation: Move to fresh air. If breathing is difficult, give oxygen. Get medical attention.
Ingestion: Do not induce vomiting. Rinse mouth, give water to drink if the person is conscious. Get medical attention immediately.

SECTION 5. FIRE-FIGHTING MEASURES

The product is not flammable. Contact with aluminium, zinc, tin and their alloys generates hydrogen gas, which is flammable. Use extinguishing media appropriate to surrounding materials. Wear self-contained breathing apparatus and full protective clothing.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Evacuate unnecessary personnel. Wear the protective equipment in Section 8. Contain the spill with inert absorbent (sand, vermiculite). Neutralise carefully with dilute acid if trained to do so. Do not allow to enter drains or waterways. Collect into a suitable corrosion-resistant container for disposal.

SECTION 7. HANDLING AND STORAGE

Handling: Avoid contact with skin, eyes and clothing. Do not breathe mist. Add caustic to water slowly, never water to caustic.
Storage: Store in a cool, dry, well-ventilated area in the original corrosion-resistant container (polyethylene, lined steel, or stainless steel). Keep away from acids, aluminium, zinc, tin, organic halogens and nitro compounds. Keep containers closed.
Solution may freeze below 54°F (12°C).

SECTION 8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Exposure limits: Sodium hydroxide — OSHA PEL 2 mg/m³ (TWA); ACGIH TLV 2 mg/m³ (ceiling); NIOSH REL 2 mg/m³ (ceiling). Engineering controls: Local exhaust ventilation where mist may be generated; emergency eyewash and safety shower within 10 seconds of work areas. Eye and face protection: Chemical splash goggles and face shield. Skin protection: Neoprene, nitrile or butyl rubber gloves, chemical-resistant apron or suit, rubber boots. Respiratory protection: Where mist exceeds exposure limits, a NIOSH-approved respirator with particulate filter.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Clear, colourless to slightly turbid liquid. Odour: Odourless. pH: 14. Freezing point: 54°F (12°C). Boiling point: 293°F (145°C). Specific gravity: 1.52 at 68°F (20°C). Solubility in water: Complete. Vapour pressure: 1.5 mm Hg at 68°F.

SECTION 10. STABILITY AND REACTIVITY

Stable under recommended storage conditions. Reacts violently with acids. Reacts with aluminium, zinc, tin, brass and bronze to produce hydrogen. Absorbs carbon dioxide from the air. Hazardous decomposition: none under normal use.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute effects: Severe burns to skin, eyes and mucous membranes on contact; ingestion causes severe burns to the mouth, throat and stomach, possibly perforation. Chronic effects: Repeated exposure to mist may cause dermatitis and damage to the respiratory tract. Carcinogenicity: Not listed by IARC, NTP or OSHA.

SECTION 12. ECOLOGICAL INFORMATION

Harmful to aquatic life through pH change. Do not release to the environment. Rapidly neutralised in the environment; does not bioaccumulate.

SECTION 13. DISPOSAL CONSIDERATIONS

Dispose of in accordance with federal, state and local regulations. Unneutralised product is a corrosive hazardous waste (RCRA D002).

SECTION 14. TRANSPORT INFORMATION

UN number: UN1824. Proper shipping name: Sodium hydroxide solution. Hazard class: 8. Packing group: II. DOT reportable quantity (RQ): 1,000 lb (sodium hydroxide).

SECTION 15. REGULATORY INFORMATION

TSCA: Listed. CERCLA RQ: 1,000 lb. SARA 311/312 hazard categories: Skin corrosion, Serious eye damage, Corrosive to metals. California Proposition 65: Not listed.

SECTION 16. OTHER INFORMATION

Prepared by: Northwest Alkali Products, Regulatory Affairs. Revision date: March 18, 2024. Changes from version 4.0: Section 8 updated to current ACGIH TLV; Section 14 packing group corrected. Disclaimer: The information in this SDS is believed to be accurate as of the revision date. It is the user's responsibility to determine suitability for their use.